



On almost bipartite non-König–Egerváry graphs

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ABSTRACT

A set $S \subseteq V$ is *independent* in a graph $G = (V, E)$ if no two vertices from S are adjacent. The *independence number* $\alpha(G)$ is the cardinality of a maximum independent set, while $\mu(G)$ is the size of a maximum matching in G . If $\alpha(G) + \mu(G)$ equals the order of G , then G is a *König–Egerváry graph* (Deming, 1979; Gavril, 1977; Sterboul, 1979). The number $d(G) = \max\{|A| - |N(A)| : A \subseteq V\}$ is the *critical difference* of G (Zhang, 1990) (where $N(A) = \{v : v \in V, N(v) \cap A \neq \emptyset\}$). It is known that the inequality $\alpha(G) - \mu(G) \leq d(G)$ holds for every graph (Levit and Mandrescu, 2012; Lorentzen, 1966; Schrijver, 2003).

A graph G is (i) *unicyclic* if it has a unique cycle, (ii) *almost bipartite* if it has only one odd cycle. Let $\ker(G) = \bigcap \{S : S \text{ is a critical independent set of } G\}$, $\text{core}(G)$ be the intersection of all maximum independent sets, and $\text{corona}(G)$ be the union of all maximum independent sets of G . It is known that $\ker(G) \subseteq \text{core}(G)$ for every graph (Levit and Mandrescu, 2012), while the equality holds for bipartite graphs (Levit and Mandrescu, 2013), and for unicyclic non-König–Egerváry graphs (Levit and Mandrescu, 2014).

In this paper, we prove that if G is an almost bipartite non-König–Egerváry graph, then $\ker(G) = \text{core}(G)$, $\text{corona}(G) \cup N(\text{core}(G)) = V(G)$, and $|\text{corona}(G)| + |\text{core}(G)| = 2\alpha(G) + 1$.

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1. Introduction

Throughout this paper $G = (V, E)$ is a finite, undirected, loopless graph without multiple edges, with vertex set $V = V(G)$ of cardinality $n(G)$, and edge set $E = E(G)$ of size $m(G)$. If $X \subset V$, then $G[X]$ is the subgraph of G spanned by X . By $G - W$ we mean the subgraph $G[V - W]$, if $W \subset V(G)$. For $F \subset E(G)$, by $G - F$ we denote the subgraph of G obtained by deleting the edges of F , and we use $G - e$, if $F = \{e\}$. If $A, B \subset V$ and $A \cap B = \emptyset$, then (A, B) stands for the set $\{e = ab : a \in A, b \in B, e \in E\}$. The neighborhood of a vertex $v \in V$ is the set $N(v) = \{w : w \in V \text{ and } vw \in E\}$, and $N(A) = \bigcup \{N(v) : v \in A\}$, $N[A] = A \cup N(A)$ for $A \subset V$. In order to avoid ambiguity, we use $N_G(v)$ instead of $N(v)$, and $N_G(A)$ instead of $N(A)$.

A *cycle* is a trail, where the only repeated vertices are the first and last ones. The graph G is *unicyclic* if it has a unique cycle. By C_n, K_n we mean the chordless cycle on $n \geq 3$ vertices, and respectively the complete graph on $n \geq 1$ vertices.

Let us define the *trace* of a family $\mathcal{F}$ of sets on the set X as $\mathcal{F}|_X = \{F \cap X : F \in \mathcal{F}\}$.

A set $S \subseteq V(G)$ is *independent* if no two vertices from S are adjacent, and by $\text{Ind}(G)$ we mean the family of all the independent sets of G . An independent set of maximum size will be referred to as a *maximum independent set*. The *independence number* of G , denoted by $\alpha(G)$, is the cardinality of a maximum independent set of G . Let $\Omega(G) = \{S : S$

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is a maximum independent set of G), $\text{core}(G) = \bigcap \{S : S \in \Omega(G)\}$ [11], and $\text{corona}(G) = \bigcup \{S : S \in \Omega(G)\}$ [2]. Clearly, $\alpha(G) \leq \alpha(G - e) \leq \alpha(G) + 1$ holds for each edge $e \in E(G)$. An edge $e \in E(G)$ is α -critical whenever $\alpha(G - e) > \alpha(G)$.

The number $d_G(X) = |X| - |N(X)|$ is the difference of the set $X \subseteq V(G)$, and $d(G) = \max\{d_G(X) : X \subseteq V\}$ is called the critical difference of G . A set $U \subseteq V(G)$ is critical if $d_G(U) = d(G)$ [26]. The number $\text{id}(G) = \max\{d_G(I) : I \in \text{Ind}(G)\}$ is called the critical independence difference of G . If $A \subseteq V(G)$ is independent and $d_G(A) = \text{id}(G)$, then A is called critical independent [26]. Clearly, $d(G) \geq \text{id}(G)$ is true for every graph G . It is known that the equality $d(G) = \text{id}(G)$ holds for every graph G [26]. For a graph G , let $\text{ker}(G) = \bigcap \{S : S \text{ is a critical independent set of } G\}$ [16]. In addition, it is worth mentioning that the inequality $\alpha(G) - \mu(G) \leq d(G)$ holds for every graph [16,21,23].

Theorem 1.1. (i) [16] $\text{ker}(G)$ is the unique minimal critical (independent) set of G , and $\text{ker}(G) \subseteq \text{core}(G)$ for every graph.
(ii) [17,18] If G is a bipartite graph, or a unicyclic non-König-Egerváry graph, then $\text{ker}(G) = \text{core}(G)$.

A matching (i.e., a set of non-incident edges of G) of maximum cardinality $\mu(G)$ is a maximum matching in G . It is well-known that $\lfloor \frac{n(G)}{2} \rfloor + 1 \leq \alpha(G) + \mu(G) \leq n(G)$ hold for every graph G . If $\alpha(G) + \mu(G) = n(G)$, then G is a König-Egerváry graph [5,7,24]. Various properties of König-Egerváry graphs are presented in [3,4,8,10,12–14]. It is known that every bipartite graph is a König-Egerváry graph [6,9]. This class includes also non-bipartite graphs (see, for instance, the graph G in Fig. 1).

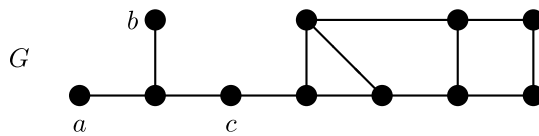


Fig. 1. G is a König-Egerváry graph with $\text{core}(G) = \{a, b, c\}$ and $\text{ker}(G) = \{a, b\}$.

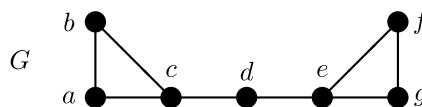


Fig. 2. $\alpha(G) = 3$, $\text{core}(G) = \{d\}$, $\text{corona}(G) = \{a, b, d, f, g\}$.

Theorem 1.2. If G is a König-Egerváry graph, then

- (i) [12] $\text{corona}(G) \cup N(\text{core}(G)) = V(G)$;
- (ii) [19] $|\text{core}(G)| + |\text{corona}(G)| = 2\alpha(G)$.

Notice that the equalities presented in Theorem 1.2 are not strong enough to assure that G is a König-Egerváry graph, e.g., see the graph G from Fig. 2. On the other hand, the butterfly graph (obtained by joining 2 copies of C_3 with a common vertex) does not satisfy Theorem 1.2(i), and the cycle graph C_5 does not satisfy Theorem 1.2(ii).

Moreover, there are graphs that do not satisfy both assertions of Theorem 1.2; see, for instance, the graph G from Fig. 3.

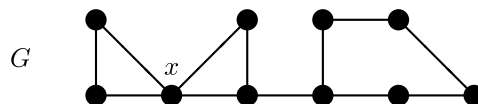


Fig. 3. $\alpha(G) = 4$, $\text{core}(G) = \emptyset$, $\text{corona}(G) = V(G) - \{x\}$.

We call a graph G almost bipartite if it has a unique odd cycle, denoted $C = (V(C), E(C))$. Since C is unique, it is chordless, and there is no other cycle of G sharing edges with C . For every $y \in V(C)$, let us define $D_y = (V_y, E_y)$ as the connected bipartite subgraph of $G - E(C)$ containing y , and

$$N_1(C) = \{v : v \in V(G) - V(C), N(v) \cap V(C) \neq \emptyset\}.$$

Clearly, every unicyclic graph with an odd cycle is almost bipartite.

Proposition 1.3. If G is almost bipartite with $C = (V(C), E(C))$ as its unique odd cycle, then $V(D_a) \cap V(D_b) = \emptyset$ for every two different vertices $a, b \in V(C)$.

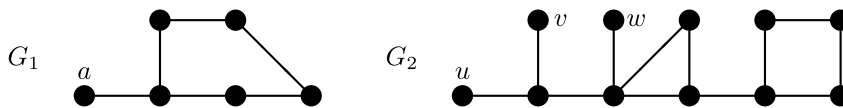


Fig. 4. Almost bipartite König-Egerváry graphs with $\ker(G_1) = \emptyset$ and $\ker(G_2) = \{u, v\}$.

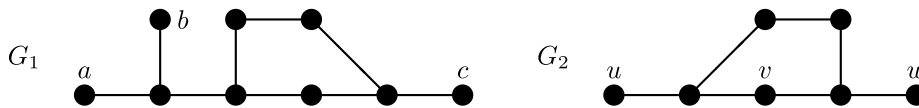


Fig. 5. G_1, G_2 are König-Egerváry graphs, $\text{core}(G_1) = \{a, b, c\}$, $\text{core}(G_2) = \{u, v, w\}$.

Proof. Assume, to the contrary, that there exist $a, b \in V(C)$, such that $V(D_a) \cap V(D_b) \neq \emptyset$. Let $x \in V(D_a) \cap V(D_b)$. Thus, there exists some path containing x , and connecting a and b . Let P_1 be a shortest one of this kind. On the other hand, there exist two paths, say P_2 and P_3 , connecting a and b , and containing only vertices belonging to C . Therefore, either P_1 and P_2 , or P_1 and P_3 , give birth to an odd cycle, different from C , and thus contradicting the fact that C is the unique odd cycle of G . ■

As a consequence of Proposition 1.3, we may infer that $\{V(D_y) : y \in V(C)\}$ is a partition of $V(G)$.

There exist König-Egerváry graphs G with $\ker(G) \neq \text{core}(G)$; for instance, the graph in Fig. 1. There are also almost bipartite König-Egerváry graph that may have $\ker(G) \neq \text{core}(G)$; e.g., the graphs in Fig. 4 have $\text{core}(G_1) = \{a\}$ and $\text{core}(G_2) = \{u, v, w\}$.

If $H_j, j = 1, 2, \dots, k$, are all the connected components of G , it is easy to see that $S \in \Omega(G)$ if and only if for every $1 \leq j \leq k$ there exists $S_j \in \Omega(H_j)$ such that $S = \bigcup_{j=1}^k S_j$. Moreover,

$$\text{core}(G) = \bigcup_{j=1}^k \text{core}(H_j), \text{ corona}(G) = \bigcup_{j=1}^k \text{corona}(H_j), \text{ and } \ker(G) = \bigcup_{j=1}^k \ker(H_j).$$

In this paper we show that for every almost bipartite non-König-Egerváry graph G , the following equalities are true:

- (i) $\ker(G) = \text{core}(G)$;
- (ii) $\text{corona}(G) \cup N(\text{core}(G)) = V(G)$;
- (iii) $|\text{corona}(G)| + |\text{core}(G)| = 2\alpha(G) + 1$.

Since $|\text{corona}(H)| + |\text{core}(H)| = 2\alpha(H)$ and the assertions (i) and (ii) hold for every bipartite connected component H of G , we may assume that every almost bipartite non-König-Egerváry graph is connected.

2. Results

Recall the following useful findings.

Lemma 2.1 ([15]). For every bipartite graph H , a vertex $v \in \text{core}(H)$ if and only if there exists a maximum matching that does not saturate v .

Lemma 2.1 fails for non-bipartite König-Egerváry graphs; e.g., every maximum matching in the graph G from Fig. 1 saturates $c \in \text{core}(G) = \{a, b, c\}$.

Lemma 2.2 ([20]). If G is an almost bipartite graph, then

- (i) $n(G) - 1 \leq \alpha(G) + \mu(G) \leq n(G)$;
- (ii) $n(G) - 1 = \alpha(G) + \mu(G)$ if and only if each edge of its unique odd cycle is α -critical.

Theorem 2.3 ([20]). If G is an almost bipartite non-König-Egerváry graph, then

- (i) $\text{core}(G) \cap N[V(C)] = \emptyset$;
- (ii) $\text{core}(G) = \bigcup_{y \in V(C)} \text{core}(D_y - y)$;
- (iii) $\Omega(G) \upharpoonright_{V(D_y - y)} = \Omega(D_y - y)$ for every $y \in V(C)$.

The assertion in Theorem 2.3 (ii) may fail for connected unicyclic König-Egerváry graphs. For instance,

$$\text{core}(G_2) \neq \{u, w\} = \bigcup_{y \in V(C)} \text{core}(D_y - y),$$

while $\text{core}(G_1) = \bigcup_{y \in V(C)} \text{core}(D_y - y)$, where G_1 and G_2 are from Fig. 5.

Proposition 2.4 ([20]). Let G be an almost bipartite graph. Then the following assertions are equivalent:

- (i) $y \in \text{core}(D_y)$, for every $y \in V(C)$;
- (ii) there exists some $S \in \Omega(G)$, such that $S \cap N_1(C) = \emptyset$;
- (iii) $n(G) - 1 = \alpha(G) + \mu(G)$, i.e., G is not a König–Egerváry graph.

Corollary 2.5. If G is an almost bipartite non-König–Egerváry graph, then there exists some $S \in \Omega(G)$, such that $|S \cap V(C)| = \lfloor \frac{|V(C)|}{2} \rfloor$, where C is its unique odd cycle.

Lemma 2.6. If G is an almost bipartite non-König–Egerváry graph, then

$$\alpha(G) = \sum_{y \in V(C)} \alpha(D_y) - \left\lfloor \frac{|V(C)|}{2} \right\rfloor - 1,$$

where C is its unique odd cycle.

Proof. By Corollary 2.5, there is a maximum independent set $S \in \Omega(G)$ such that $|S \cap V(C)| = \lfloor \frac{|V(C)|}{2} \rfloor$. Therefore, by Proposition 2.4 (i),

$$\begin{aligned} \alpha(G) &= \sum_{y \in S \cap V(C)} \alpha(D_y) + \sum_{y \in V(C) - S} (\alpha(D_y) - 1) \\ &= \sum_{y \in V(C)} \alpha(D_y) - |V(C) - S| = \sum_{y \in V(C)} \alpha(D_y) - \left\lfloor \frac{|V(C)|}{2} \right\rfloor - 1, \end{aligned}$$

as required. ■

Proposition 2.7. If G is an almost bipartite non-König–Egerváry graph, then every maximum matching in G contains at least one edge belonging to its unique odd cycle.

Proof. Assume, to the contrary, that there exists some maximum matching M in G , such that $M \cap E(C) = \emptyset$.

Case 1. There exist two consecutive vertices on C , say y_1, y_2 , such that $D_{y_1} = \{y_1\}$ and $D_{y_2} = \{y_2\}$.

Since $G - y_1y_2$ is a bipartite graph, we have that

$$\begin{aligned} \alpha(G) + \mu(G) + 1 &= n(G) = n(G - y_1y_2) \\ &= \alpha(G - y_1y_2) + \mu(G - y_1y_2) = \alpha(G) + 1 + \mu(G - y_1y_2) \end{aligned}$$

which leads to $\mu(G - y_1y_2) = \mu(G) = |M|$. Since $M \cap E(C) = \emptyset$, we infer that $M \cup \{y_1y_2\}$ is a matching in G , larger than M , contradicting the fact that $\mu(G) = |M|$.

Case 2. No two consecutive vertices on C , say y_1, y_2 , satisfy both $D_{y_1} = \{y_1\}$ and $D_{y_2} = \{y_2\}$. It follows that the number k of vertices $y_1, y_2, \dots, y_k$ on C with $D_{y_i} = \{y_i\}$ satisfies $k \leq \lfloor \frac{|V(C)|}{2} \rfloor$.

Let $y_{k+1}, y_{k+2}, \dots, y_{k+p}$ be all the vertices on C with $|V(D_{y_i})| = n(D_{y_i}) \geq 2$. Hence, $p \geq \lfloor \frac{|V(C)|}{2} \rfloor$. Since every D_{y_i} is bipartite, we know that $n(D_{y_i}) = \alpha(D_{y_i}) + \mu(D_{y_i})$. In addition, $\mu(G) = \sum_{i=k+1}^{k+p} \mu(D_{y_i})$, because $M \cap E(C) = \emptyset$. Thus

$$n(G) = \sum_{i=1}^{k+p} n(D_{y_i}) = \sum_{i=1}^k n(D_{y_i}) + \sum_{i=k+1}^{k+p} n(D_{y_i}) = k + \sum_{i=k+1}^{k+p} n(D_{y_i}).$$

Consequently, by Proposition 2.4 (iii) and Lemma 2.6,

$$\begin{aligned} n(G) &= \alpha(G) + \mu(G) + 1 = \sum_{y \in V(C)} \alpha(D_y) - \left\lfloor \frac{|V(C)|}{2} \right\rfloor - 1 + \mu(G) + 1 \\ &= \sum_{y \in V(C)} \alpha(D_y) - \left\lfloor \frac{|V(C)|}{2} \right\rfloor + \mu(G). \end{aligned}$$

On the other hand, we have

$$n(G) = k + \sum_{i=k+1}^{k+p} n(D_{y_i}) = k + \sum_{i=k+1}^{k+p} \alpha(D_{y_i}) + \sum_{i=k+1}^{k+p} \mu(D_{y_i}) = k + \sum_{i=k+1}^{k+p} \alpha(D_{y_i}) + \mu(G).$$

Hence, we get

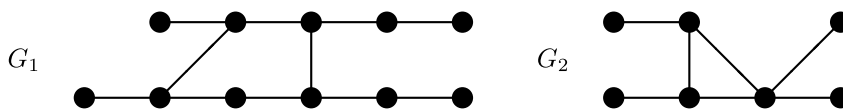


Fig. 6. G_1 and G_2 are almost bipartite König-Egerváry graphs.

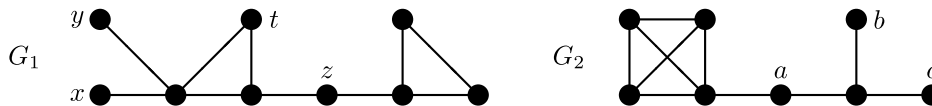


Fig. 7. $\ker(G_1) = \{x, y\}$, $\ker(G_2) = \{b, c\}$ and only G_1 is a König-Egerváry graph.

$$\sum_{y \in V(C)} \alpha(D_y) - \left\lfloor \frac{|V(C)|}{2} \right\rfloor = k + \sum_{i=k+1}^{k+p} \alpha(D_{y_i})$$

$$\sum_{i=1}^k \alpha(D_{y_i}) = k + \left\lfloor \frac{|V(C)|}{2} \right\rfloor.$$

Taking into account that $\sum_{i=1}^k \alpha(D_{y_i}) = k$ by definition of the sequence $y_1, y_2, \dots, y_k$, we arrive at a contradiction. ■

Proposition 2.7 is not true for almost bipartite König-Egerváry graphs; e.g., see G_2 in Fig. 6.

Lemma 2.8 ([20]). *Let G be an almost bipartite non-König-Egerváry graph with the unique odd cycle C .*

- (i) *If A is a critical independent set of G , then $A \cap V(C) = \emptyset$.*
- (ii) *$\text{core}(G)$ is a critical set.*

Lemma 2.9. *Let G be an almost bipartite graph. If there is $x \in N_1(C)$, such that $x \in \text{core}(D_y - y)$ for some $y \in V(C)$, then G is a König-Egerváry graph.*

Proof. Let $x \in \text{core}(D_y - y)$, $y \in N(x) \cap V(C)$, and $z \in N(y) \cap V(C)$. Suppose, to the contrary, that G is not a König-Egerváry graph. By Lemma 2.2, the edge yz is α -critical. By Lemma 2.8, $y \notin \text{core}(G)$. Thus it follows that $\alpha(G) = \alpha(G - y)$. By Lemma 2.1 there exists a maximum matching M_x of $D_y - y$ not saturating x . Combining M_x with a maximum matching in $G - D_y$ we get a maximum matching M_y of $G - y$. Hence $M_y \cup \{xy\}$ is a matching in G , which results in $\mu(G) \geq \mu(G - y) + 1$. Consequently, using Lemma 2.2(ii) and having in mind that $G - y$ is a bipartite graph of order $n(G) - 1$, we get the following contradiction

$$n(G) - 1 = \alpha(G) + \mu(G) \geq \alpha(G - y) + \mu(G - y) + 1 = n(G) - 1 + 1 = n(G),$$

and this completes the proof. ■

There exist König-Egerváry and non-König-Egerváry graphs having $\text{core}(G) \neq \ker(G)$; e.g., the graphs from Fig. 7: $\text{core}(G_1) = \{x, y, z, t\}$ and $\text{core}(G_2) = \{a, b, c\}$.

Theorem 2.10. *Let G be an almost bipartite non-König-Egerváry graph with the unique odd cycle C . Then*

$$\ker(G) = \bigcup_{y \in V(C)} \ker(D_y - y) = \bigcup_{y \in V(C)} \text{core}(D_y - y) = \text{core}(G).$$

Proof. By Theorem 2.3, we have that $\text{core}(G) = \bigcup_{y \in V(C)} \text{core}(D_y - y)$.

Since every $D_y - y$ is a bipartite graph, we infer that $\ker(D_y - y) = \text{core}(D_y - y)$, by Theorem 1.1(ii). Consequently, we obtain

$$\text{core}(G) = \bigcup_{y \in V(C)} \text{core}(D_y - y) = \bigcup_{y \in V(C)} \ker(D_y - y).$$

By Lemma 2.8(ii), the set $\text{core}(G)$ is critical in G . Hence, we get that

$$\ker(G) \subseteq \text{core}(G) = \bigcup_{y \in V(C)} \ker(D_y - y).$$

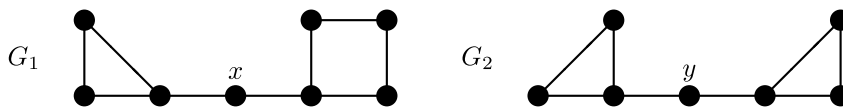


Fig. 8. $\text{core}(G_1) = \{x\}$, $\text{core}(G_2) = \{y\}$ and only G_1 is a König-Egerváry graph.

Thus it is enough to show that

$$\bigcup_{y \in V(C)} \ker(D_y - y) \subseteq \ker(G).$$

By Lemma 2.9, if $A \subseteq \ker(D_y - y)$, then $N_G(A) = N_{D_y - y}(A)$, since G is a non-König-Egerváry almost bipartite graph. Hence it follows $d_G(A) = d_{D_y - y}(A)$ for every $A \subseteq \ker(D_y - y)$. Thus, in accordance with Theorem 1.1(i), if $A \subset \ker(D_y - y)$, then

$$d_G(A) = d_{D_y - y}(A) < d_{D_y - y}(\ker(D_y - y)) = d_G(\ker(D_y - y)). \quad (*)$$

Since $\ker(G) \subseteq \bigcup_{y \in V(C)} \ker(D_y - y)$,

$$\begin{aligned} d_G(\ker(G)) &= d_G\left(\ker(G) \cap \bigcup_{y \in V(C)} \ker(D_y - y)\right) \\ &= d_G\left(\bigcup_{y \in V(C)} (\ker(D_y - y) \cap \ker(G))\right) = \sum_{y \in V(C)} d_G(\ker(D_y - y) \cap \ker(G)). \end{aligned}$$

If $\ker(G) \neq \bigcup_{y \in V(C)} \ker(D_y - y)$, then

$$\ker(G) \upharpoonright_{V(D_y - y)} \subset \ker(D_y - y)$$

for some $y \in V(C)$. Consequently, using the inequality (*) for $A = \ker(G) \upharpoonright_{V(D_y - y)}$, we obtain

$$\begin{aligned} d_G(\ker(G)) &= \sum_{y \in V(C)} d_G(\ker(G) \upharpoonright_{V(D_y - y)}) \\ &< \sum_{y \in V(C)} d_G(\ker(D_y - y)) = d_G\left(\bigcup_{y \in V(C)} \ker(D_y - y)\right) = d(\text{core}(G)) = d(G), \end{aligned}$$

which stays in contradiction with the fact that $\ker(G)$ is critical in G . ■

As a consequence, we get the following.

Corollary 2.11 ([18]). *If G is a unicyclic non-König-Egerváry graph, then $\ker(G) = \text{core}(G)$.*

It is easy to see that for every non-negative integer k there exists a graph G with $|\text{core}(G)| = k$. For instance, $|\text{core}(K_3)| = 0$, while the graph G , obtained by joining $k \geq 1$ leaves to one of the vertices of K_3 , has $|\text{core}(G)| = k$.

The proof of Corollary 2.6 from [18] contains the following general finding.

Proposition 2.12. *If G is a graph without isolated vertices, then $|\ker(G)| \neq 1$.*

Combining Theorem 2.10 with Proposition 2.12, and the assumption made in Introduction that almost bipartite non-König-Egerváry graphs are connected, we obtain the following.

Corollary 2.13. *If G is an almost bipartite non-König-Egerváry graph, then $|\text{core}(G)| \neq 1$.*

Corollary 2.14 ([18]). *If G is a unicyclic non-König-Egerváry graph, then $|\text{core}(G)| \neq 1$.*

There exist non-bipartite König-Egerváry graphs and non-König-Egerváry graphs that have $|\text{core}(G)| = 1$; e.g., the graph G_1 in Fig. 4 and the graphs in Fig. 8.

It is worth noticing that there exists an almost bipartite König-Egerváry graph with a critical independent set meeting its unique cycle. For instance, the bull graph.

There exist non-König-Egerváry graphs satisfying $\text{corona}(G) \cup N(\text{core}(G)) \neq V(G)$; e.g., the graph in Fig. 9 has $\text{corona}(G) \cup N(\text{core}(G)) = V(G) - \{a\}$.

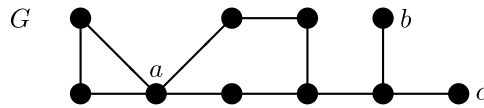


Fig. 9. G is a non-König-Egerváry graph with $\text{core}(G) = \{b, c\}$.

Theorem 2.15. *If G is an almost bipartite non-König-Egerváry graph, then*

- (i) $\text{corona}(G) \cup N(\text{core}(G)) = V(G)$;
- (ii) $\text{corona}(G) = V(C) \cup \left(\bigcup_{y \in V(C)} \text{corona}(D_y - y) \right)$.

Proof. (i) It is enough to show that $V(G) \subseteq \text{corona}(G) \cup N(\text{core}(G))$.

Let $a \in V(G)$.

Case 1. $a \in V(C)$. If $b \in N(a) \cap V(C)$, then, by Lemma 2.2(ii), the edge ab is α -critical. Hence $a \in \text{corona}(G)$.

Case 2. $a \in V(G) - V(C)$. It follows that $a \in V(D_y - y)$, for some $y \in V(C)$.

Since $G[D_y - y]$ is bipartite, by Theorem 1.2 (iii), we know that $V(D_y - y) = \text{corona}(D_y - y) \cup N(\text{core}(D_y - y))$, while by Theorem 2.3(iii), we have that $\Omega(G)|_{V(D_y - y)} = \Omega(D_y - y)$ for every $y \in V(C)$, which ensures that $\text{corona}(D_y - y) \subseteq \text{corona}(G)$.

Therefore, either $a \in \text{corona}(D_y - y) \subseteq \text{corona}(G)$, or $a \in N(\text{core}(D_y - y)) \subseteq N(\text{core}(G))$, because $\text{core}(D_y - y) \subseteq \text{core}(G)$, by Theorem 2.3(ii). Thus, $a \in \text{corona}(G) \cup N(\text{core}(G))$.

All in all, $V(G) = \text{corona}(G) \cup N(\text{core}(G))$.

(ii) In the proof of Part (i) we showed that $\text{corona}(D_y - y) \subseteq \text{corona}(G)$ for every $y \in V(C)$, and $V(C) \subseteq \text{corona}(G)$.

Hence, $V(C) \cup \left(\bigcup_{y \in V(C)} \text{corona}(D_y - y) \right) \subseteq \text{corona}(G)$. To complete the proof, it remains to validate that $\text{corona}(G) \subseteq V(C) \cup \left(\bigcup_{y \in V(C)} \text{corona}(D_y - y) \right)$. Let $a \in \text{corona}(G)$. Then, $a \in S$ for some $S \in \Omega(G)$. Suppose $a \notin V(C)$, then there must be $y \in V(C)$ such that $a \in D_y - y$. Thus, $a \in S \cap V(D_y - y) \subseteq \text{corona}(D_y - y)$, because $\Omega(G)|_{V(D_y - y)} = \Omega(D_y - y)$, in accordance with Theorem 2.3(iii). ■

Theorem 2.16 ([20]). *If G is an almost bipartite non-König-Egerváry graph, then*

$$d(G) = \alpha(G) - \mu(G) = |\text{core}(G)| - |N(\text{core}(G))|.$$

Let us mention that for unicyclic non-König-Egerváry graphs the equality $d(G) = \alpha(G) - \mu(G)$ was proved in [1]. A characterization of graphs satisfying all the equalities in Theorem 2.16 is presented in [22].

Theorem 2.17. *If G is an almost bipartite non-König-Egerváry graph, then*

$$|\text{corona}(G)| + |\text{core}(G)| = 2\alpha(G) + 1.$$

Proof. Let $S \in \Omega(G)$. According to Theorem 2.15 and Lemma 2.2, we infer that

$$|\text{corona}(G)| + |N(\text{core}(G))| = |V(G)| = \alpha(G) + \mu(G) + 1.$$

By Theorem 2.16, we obtain

$$\begin{aligned} |\text{corona}(G)| + |\text{core}(G)| &= |\text{corona}(G)| + |N(\text{core}(G))| + \alpha(G) - \mu(G) \\ &= \alpha(G) + \mu(G) + 1 + \alpha(G) - \mu(G) = 2\alpha(G) + 1 \end{aligned}$$

as required. ■

Corollary 2.18 ([18]). *If G is a unicyclic non-König-Egerváry graph, then $|\text{corona}(G)| + |\text{core}(G)| = 2\alpha(G) + 1$.*

3. Conclusions

It is known that $\ker(G) \subseteq \text{core}(G)$ for every graph G [16]. By Theorem 2.10, each almost bipartite non-König-Egerváry graph satisfies $\ker(G) = \text{core}(G)$, like bipartite graphs [17] and unicyclic non-König-Egerváry graphs [18].

Problem 3.1. Characterize graphs enjoying $\ker(G) = \text{core}(G)$.

Let us mention that [Problem 3.1](#) has been solved for König-Egerváry graphs in [25].

[Theorem 2.15](#) asserts that $\text{corona}(G) \cup N(\text{core}(G)) = V(G)$ is true for almost bipartite non-König-Egerváry graphs, like for König-Egerváry graphs [12] and unicyclic non-König-Egerváry graphs [18].

Problem 3.2. Characterize graphs enjoying $\text{corona}(G) \cup N(\text{core}(G)) = V(G)$.

[Theorem 2.17](#) claims that $|\text{corona}(G)| + |\text{core}(G)| = 2\alpha(G) + 1$ holds for almost bipartite non-König-Egerváry graphs, like for unicyclic non-König-Egerváry graphs [18].

Problem 3.3. Characterize graphs enjoying $|\text{corona}(G)| + |\text{core}(G)| = 2\alpha(G) + 1$.

[Proposition 2.7](#) motivates the following.

Conjecture 3.4. If G is an almost bipartite non-König-Egerváry graph, then every maximum matching in G contains $\lfloor \frac{|V(C)|}{2} \rfloor$ edges belonging to its unique odd cycle C .

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Data availability

No data was used for the research described in the article.

References

- [1] A. Bhattacharya, A. Mondal, T.S. Murthy, Problems on matchings and independent sets of a graph, *Discrete Math.* 341 (2018) 1561–1572, <http://dx.doi.org/10.1016/j.disc.2018.02.021>.
- [2] E. Boros, M.C. Golumbic, V.E. Levit, On the number of vertices belonging to all maximum stable sets of a graph, *Discrete Appl. Math.* 124 (2002) 17–25, [http://dx.doi.org/10.1016/S0166-218X\(01\)00327-4](http://dx.doi.org/10.1016/S0166-218X(01)00327-4).
- [3] J.M. Bourjolly, P.L. Hammer, B. Simeone, Node weighted graphs having König-Egerváry property, *Math. Program. Study* 22 (1984) 44–63, <http://dx.doi.org/10.1007/BFb0121007>.
- [4] J.M. Bourjolly, W.R. Pulleyblank, König-egerváry graphs, 2-bicritical graphs and fractional matchings, *Discrete Appl. Math.* 24 (1989) 63–82, [http://dx.doi.org/10.1016/0166-218X\(92\)90273-D](http://dx.doi.org/10.1016/0166-218X(92)90273-D).
- [5] R.W. Deming, Independence numbers of graphs – an extension of the König-Egerváry theorem, *Discrete Math.* 27 (1979) 23–33, [http://dx.doi.org/10.1016/0012-365X\(79\)90066-9](http://dx.doi.org/10.1016/0012-365X(79)90066-9).
- [6] E. Egerváry, On combinatorial properties of matrices, *Mat. Lapok* 38 (1931) 16–28.
- [7] F. Gavril, Testing for equality between maximum matching and minimum node covering, *Inform. Process. Lett.* 6 (1977) 199–202, [http://dx.doi.org/10.1016/0020-0190\(77\)90068-0](http://dx.doi.org/10.1016/0020-0190(77)90068-0).
- [8] A. Jarden, V.E. Levit, E. Mandrescu, Two more characterizations of König-Egerváry graphs, *Discrete Appl. Math.* 231 (2017) 175–180, <http://dx.doi.org/10.1016/j.dam.2016.05.012>.
- [9] D. König, Graphen und matrizen, *Mat. Lapok* 38 (1931) 116–119.
- [10] V.E. Levit, E. Mandrescu, On the structure of α -stable graphs, *Discrete Math.* 236 (2001) 227–243, [http://dx.doi.org/10.1016/S0012-365X\(00\)00444-1](http://dx.doi.org/10.1016/S0012-365X(00)00444-1).
- [11] V.E. Levit, E. Mandrescu, Combinatorial properties of the family of maximum stable sets of a graph, *Discrete Appl. Math.* 117 (2002) 149–161, [http://dx.doi.org/10.1016/S0166-218X\(01\)00183-4](http://dx.doi.org/10.1016/S0166-218X(01)00183-4).
- [12] V.E. Levit, E. Mandrescu, On α^+ -stable König-Egerváry graphs, *Discrete Math.* 263 (2003) 179–190, [http://dx.doi.org/10.1016/S0012-365X\(02\)00528-9](http://dx.doi.org/10.1016/S0012-365X(02)00528-9).
- [13] V.E. Levit, E. Mandrescu, On α -critical edges in König-Egerváry graphs, *Discrete Math.* 306 (2006) 1684–1693, <http://dx.doi.org/10.1016/j.disc.2006.05.001>.
- [14] V.E. Levit, E. Mandrescu, Critical independent sets and König-Egerváry graphs, *Graphs Combin.* 28 (2012) 243–250, <http://dx.doi.org/10.1007/s00373-011-1037-y>.
- [15] V.E. Levit, E. Mandrescu, On the core of a unicyclic graph, *Ars Math. Contemp.* 5 (2012) 321–327, <http://dx.doi.org/10.26493/1855-3974.201.6e1>.
- [16] V.E. Levit, E. Mandrescu, Vertices belonging to all critical independent sets of a graph, *SIAM J. Discrete Math.* 26 (2012) 399–403, <http://dx.doi.org/10.1137/110823560>.
- [17] V.E. Levit, E. Mandrescu, Critical sets in bipartite graphs, *Ann. Comb.* 17 (2013) 543–548, <http://dx.doi.org/10.1007/s00026-013-0195-4>.
- [18] V.E. Levit, E. Mandrescu, On the intersection of all critical sets of a unicyclic graph, *Discrete Appl. Math.* 162 (2014) 409–414, <http://dx.doi.org/10.1016/j.dam.2013.09.006>.
- [19] V.E. Levit, E. Mandrescu, A set and collection lemma, *Electron. J. Combin.* 21 (2014) <http://dx.doi.org/10.37236/2514>, #P1.40.
- [20] V.E. Levit, E. Mandrescu, On the critical difference of almost bipartite graphs, *J. Algebraic Combin.* 56 (2022) 59–73, <http://dx.doi.org/10.1007/s10801-020-00968-x>.
- [21] L.C. Lorentzen, Notes on Covering of Arcs By Nodes in an Undirected Graph, Technical Report ORC 66-16, Operations Research Center, University of California, Berkeley, CA, 1966.
- [22] H. Lu, Z. Yang, On critical difference, independence number and matching number of graphs, *Graphs Combin.* 39 (2023) <http://dx.doi.org/10.1007/s00373-023-02690-4>, #93.
- [23] A. Schrijver, *Combinatorial Optimization*, Springer, Berlin, 2003, ISBN 3-540-44389-4.
- [24] F. Sterboul, A characterization of the graphs in which the transversal number equals the matching number, *J. Combin. Theory Ser. B* 27 (1979) 228–229, [http://dx.doi.org/10.1016/0095-8956\(79\)90085-6](http://dx.doi.org/10.1016/0095-8956(79)90085-6).
- [25] Z. Yang, H. Lu, Q. Yu, Critical independent sets of König-Egerváry graphs, *Discrete Appl. Math.* 318 (2022) 1–5, <http://dx.doi.org/10.1016/j.dam.2022.04.014>.
- [26] C.Q. Zhang, Finding critical independent sets and critical vertex subsets are polynomial problems, *SIAM J. Discrete Math.* 3 (1990) 431–438, <http://dx.doi.org/10.1137/0403037>.